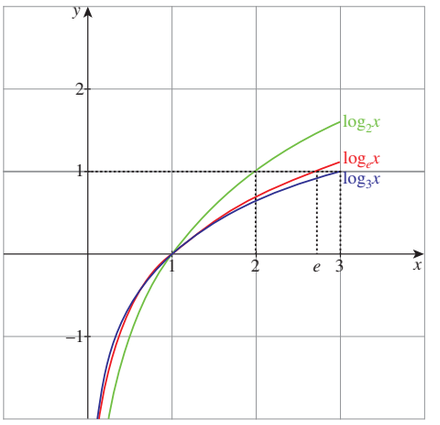


Figure 1.44

(vi) If f is a real valued function such that $f(x) \neq 0$, then the real valued function g defined by $g(x) = \frac{1}{f(x)}$ on a suitable domain is called the ***reciprocal function*** of f. The domain of g is the set obtained from $\mathbb{R}$ by removing the real numbers at which $f(x) = 0$. For example, the largest possible domain of $f(x) = \frac{1}{x-1}$ is $\mathbb{R} - \{1\}$.

Let us see two more categories of functions.

Definition 1.12

A function $f : \mathbb{R} \to \mathbb{R}$ is said to be an ***odd function*** if $f(-x) = -f(x)$ for all $x \in \mathbb{R}$. It is said to be an ***even function*** if $f(-x) = f(x)$ for all $x \in \mathbb{R}$. [See, Figures 1.45 and 1.46].

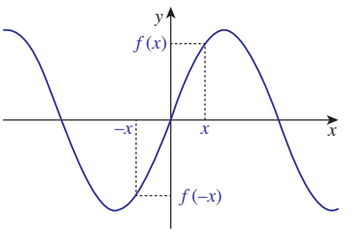


Figure 1.45

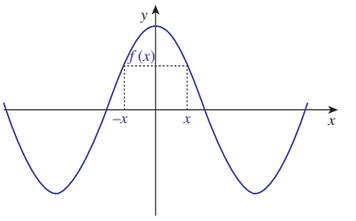


Figure 1.46

The function defined by $f(x) = x$, $f(x) = 2x$ and $f(x) = x^3 + 2x$ are some examples for odd functions. The functions defined by $f(x) = x^2$, $f(x) = 3$, $f(x) = x^4 + x^2$ and $f(x) = |x|$ are some examples for even functions. Note that the function $f(x) = x + x^2$ is neither even nor odd.

We can prove the following results.

(i) The sum of two odd functions is an odd function.
(ii) The sum of two even functions is an even function
(iii) The product of two odd functions is an even function.
(iv) The product of two even functions is an even function.
(v) The product of an odd function and an even function is an odd function.
(vi) The only function which is both odd and even function is the zero function.
(vii) The product of a positive constant and an even function is an even function.
(viii) The product of a negative constant and an even function is also an even function.
(ix) The product of a constant and an odd function is an odd function.
(x) There are functions which are neither odd nor even.

Let us prove one of the above properties. The other properties can be proved similarly.